

$$x^T y = \sum_{i=1}^n x_i y_i$$

is a scalar number (actually a 1×1 matrix) called the *inner product* of x and y . We also use the notation $\langle x, y \rangle$ to denote $x^T y$. The inner-product operator is commutative: $\langle x, y \rangle = \langle y, x \rangle$. The matrix xy^T is an $n \times n$ matrix Z called the *outer product* of x and y , where $z_{ij} = x_i y_j$. The (*euclidean*) *norm* $\|x\|$ of an n -vector x is defined by

$$\begin{aligned} \|x\| &= (x_1^2 + x_2^2 + \cdots + x_n^2)^{1/2} \\ &= (x^T x)^{1/2}. \end{aligned}$$

Thus, the norm of x is its length in n -dimensional euclidean space. A useful fact, which follows from the equality

$$((ax_1)^2 + (ax_2)^2 + \cdots + (ax_n)^2)^{1/2} = |a| (x_1^2 + x_2^2 + \cdots + x_n^2)^{1/2}$$

is that for any real number a and n -vector x ,

$$\|ax\| = |a| \|x\|. \quad (\text{D.3})$$

Exercises

D.1-1

Show that if A and B are symmetric $n \times n$ matrices, then so are $A + B$ and $A - B$.

D.1-2

Prove that $(AB)^T = B^T A^T$ and that $A^T A$ is always a symmetric matrix.

D.1-3

Prove that the product of two lower-triangular matrices is lower-triangular.

D.1-4

Prove that if P is an $n \times n$ permutation matrix and A is an $n \times n$ matrix, then the matrix product PA is A with its rows permuted, and the matrix product AP is A with its columns permuted. Prove that the product of two permutation matrices is a permutation matrix.

D.2 Basic matrix properties

We now define some basic properties pertaining to matrices: inverses, linear dependence and independence, rank, and determinants. We also define the class of positive-definite matrices.

Matrix inverses, ranks, and determinants

The **inverse** of an $n \times n$ matrix A is the $n \times n$ matrix, denoted A^{-1} (if it exists), such that $AA^{-1} = I_n = A^{-1}A$. For example,

$$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}.$$

Many nonzero $n \times n$ matrices do not have inverses. A matrix without an inverse is called **noninvertible**, or **singular**. An example of a nonzero singular matrix is

$$\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}.$$

If a matrix has an inverse, it is called **invertible**, or **nonsingular**. Matrix inverses, when they exist, are unique. (See Exercise D.2-1.) If A and B are nonsingular $n \times n$ matrices, then

$$(BA)^{-1} = A^{-1}B^{-1}.$$

The inverse operation commutes with the transpose operation:

$$(A^{-1})^T = (A^T)^{-1}.$$

The vectors $x_1, x_2, \dots, x_n$ are **linearly dependent** if there exist coefficients $c_1, c_2, \dots, c_n$, not all of which are 0, such that $c_1x_1 + c_2x_2 + \dots + c_nx_n = 0$. The row vectors $x_1 = (1 \ 2 \ 3)$, $x_2 = (2 \ 6 \ 4)$, and $x_3 = (4 \ 11 \ 9)$ are linearly dependent, for example, since $2x_1 + 3x_2 - 2x_3 = 0$. If vectors are not linearly dependent, they are **linearly independent**. For example, the columns of an identity matrix are linearly independent.

The **column rank** of a nonzero $m \times n$ matrix A is the size of the largest set of linearly independent columns of A . Similarly, the **row rank** of A is the size of the largest set of linearly independent rows of A . A fundamental property of any matrix A is that its row rank always equals its column rank, so that we can simply refer to the **rank** of A . The rank of an $m \times n$ matrix is an integer between 0 and $\min\{m, n\}$, inclusive. (The rank of a zero matrix is 0, and the rank of an $n \times n$ identity matrix is n .) An alternate, but equivalent and often more useful, definition is that the rank of a nonzero $m \times n$ matrix A is the smallest number r such that there exist matrices B and C of respective sizes $m \times r$ and $r \times n$ such that $A = BC$. A square $n \times n$ matrix has **full rank** if its rank is n . An $m \times n$ matrix has **full column rank** if its rank is n . The following theorem gives a fundamental property of ranks.

Theorem D.1

A square matrix has full rank if and only if it is nonsingular. ■

A **null vector** for a matrix A is a nonzero vector x such that $Ax = 0$. The following theorem (whose proof is left as Exercise D.2-7) and its corollary relate the notions of column rank and singularity to null vectors.

Theorem D.2

A matrix has full column rank if and only if it does not have a null vector. ■

Corollary D.3

A square matrix is singular if and only if it has a null vector. ■

The ij th **minor** of an $n \times n$ matrix A , for $n > 1$, is the $(n-1) \times (n-1)$ matrix $A_{[ij]}$ obtained by deleting the i th row and j th column of A . The **determinant** of an $n \times n$ matrix A is defined recursively in terms of its minors by

$$\det(A) = \begin{cases} a_{11} & \text{if } n = 1, \\ \sum_{j=1}^n (-1)^{1+j} a_{1j} \det(A_{[1j]}) & \text{if } n > 1. \end{cases}$$

The term $(-1)^{i+j} \det(A_{[ij]})$ is known as the **cofactor** of the element a_{ij} .

The following theorems, whose proofs are omitted, express fundamental properties of the determinant.

Theorem D.4 (Determinant properties)

The determinant of a square matrix A has the following properties:

- If any row or any column of A is zero, then $\det(A) = 0$.
- The determinant of A is multiplied by λ if the entries of any one row (or any one column) of A are all multiplied by λ .
- The determinant of A is unchanged if the entries in one row (respectively, column) are added to those in another row (respectively, column).
- The determinant of A equals the determinant of A^T .
- The determinant of A is multiplied by -1 if any two rows (or any two columns) are exchanged.

Also, for any square matrices A and B , we have $\det(AB) = \det(A) \det(B)$. ■

Theorem D.5

An $n \times n$ matrix A is singular if and only if $\det(A) = 0$. ■

Positive-definite matrices

Positive-definite matrices play an important role in many applications. An $n \times n$ matrix A is **positive-definite** if $x^T A x > 0$ for all n -vectors $x \neq 0$. For example, the identity matrix is positive-definite, since if $x = (x_1 \ x_2 \ \cdots \ x_n)^T$ is a nonzero vector, then

$$\begin{aligned} x^T I_n x &= x^T x \\ &= \sum_{i=1}^n x_i^2 \\ &> 0. \end{aligned}$$

Matrices that arise in applications are often positive-definite due to the following theorem.

Theorem D.6

For any matrix A with full column rank, the matrix $A^T A$ is positive-definite.

Proof We must show that $x^T (A^T A) x > 0$ for any nonzero vector x . For any vector x ,

$$\begin{aligned} x^T (A^T A) x &= (Ax)^T (Ax) \quad (\text{by Exercise D.1-2}) \\ &= \|Ax\|^2. \end{aligned}$$

The value $\|Ax\|^2$ is just the sum of the squares of the elements of the vector Ax . Therefore, $\|Ax\|^2 \geq 0$. We'll show by contradiction that $\|Ax\|^2 > 0$. Suppose that $\|Ax\|^2 = 0$. Then, every element of Ax is 0, which is to say $Ax = 0$. Since A has full column rank, Theorem D.2 says that $x = 0$, which contradicts the requirement that x is nonzero. Hence, $A^T A$ is positive-definite. ■

Section 28.3 explores other properties of positive-definite matrices. Section 33.3 uses a similar condition, known as positive-semidefinite. An $n \times n$ matrix A is **positive-semidefinite** if $x^T A x \geq 0$ for all n -vectors $x \neq 0$.

Exercises

D.2-1

Prove that matrix inverses are unique, that is, if B and C are inverses of A , then $B = C$.

D.2-2

Prove that the determinant of a lower-triangular or upper-triangular matrix is equal to the product of its diagonal elements. Prove that the inverse of a lower-triangular matrix, if it exists, is lower-triangular.

D.2-3

Prove that if P is a permutation matrix, then P is invertible, its inverse is P^T , and P^T is a permutation matrix.

D.2-4

Let A and B be $n \times n$ matrices such that $AB = I$. Prove that if A' is obtained from A by adding row j into row i , where $i \neq j$, then subtracting column i from column j of B yields the inverse B' of A' .

D.2-5

Let A be a nonsingular $n \times n$ matrix with complex entries. Show that every entry of A^{-1} is real if and only if every entry of A is real.

D.2-6

Show that if A is a nonsingular, symmetric, $n \times n$ matrix, then A^{-1} is symmetric. Show that if B is an arbitrary $m \times n$ matrix, then the $m \times m$ matrix given by the product BAB^T is symmetric.

D.2-7

Prove Theorem D.2. That is, show that a matrix A has full column rank if and only if $Ax = 0$ implies $x = 0$. (*Hint*: Express the linear dependence of one column on the others as a matrix-vector equation.)

D.2-8

Prove that for any two compatible matrices A and B ,

$$\text{rank}(AB) \leq \min \{\text{rank}(A), \text{rank}(B)\},$$

where equality holds if either A or B is a nonsingular square matrix. (*Hint*: Use the alternate definition of the rank of a matrix.)

Problems
D-1 Vandermonde matrix

Given numbers $x_0, x_1, \dots, x_{n-1}$, prove that the determinant of the *Vandermonde matrix*

$$V(x_0, x_1, \dots, x_{n-1}) = \begin{pmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^{n-1} \\ 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n-1} & x_{n-1}^2 & \cdots & x_{n-1}^{n-1} \end{pmatrix}$$

is

$$\det(V(x_0, x_1, \dots, x_{n-1})) = \prod_{0 \leq j < k \leq n-1} (x_k - x_j) .$$

(Hint: Multiply column i by $-x_0$ and add it to column $i + 1$ for $i = n - 1, n - 2, \dots, 1$, and then use induction.)

D-2 Permutations defined by matrix-vector multiplication over $GF(2)$

One class of permutations of the integers in the set $S_n = \{0, 1, 2, \dots, 2^n - 1\}$ is defined by matrix multiplication over $GF(2)$, the Galois field of two elements. For each integer $x \in S_n$, we view its binary representation as an n -bit vector

$$\begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ \vdots \\ x_{n-1} \end{pmatrix} ,$$

where $x = \sum_{i=0}^{n-1} x_i 2^i$. If A is an $n \times n$ matrix in which each entry is either 0 or 1, then we can define a permutation mapping each value $x \in S_n$ to the number whose binary representation is the matrix-vector product Ax . All this arithmetic is performed over **$GF(2)$** : all values are either 0 or 1, and with one exception, the usual rules of addition and multiplication apply. The exception is that $1 + 1 = 0$. You can think of arithmetic over $GF(2)$ as being just like regular integer arithmetic, except that you use only the least-significant bit.

As an example, for $S_2 = \{0, 1, 2, 3\}$, the matrix

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

defines the following permutation π_A : $\pi_A(0) = 0$, $\pi_A(1) = 3$, $\pi_A(2) = 2$, $\pi_A(3) = 1$. To see why $\pi_A(3) = 1$, observe that, working in $GF(2)$,

$$\begin{aligned}
 \pi_A(3) &= \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\
 &= \begin{pmatrix} 1 \cdot 1 + 0 \cdot 1 \\ 1 \cdot 1 + 1 \cdot 1 \end{pmatrix} \\
 &= \begin{pmatrix} 1 \\ 0 \end{pmatrix},
 \end{aligned}$$

which is the binary representation of 1.

For the remainder of this problem, we'll work over $GF(2)$, and all matrix and vector entries will be 0 or 1. Define the **rank** of a 0-1 matrix (a matrix for which each entry is either 0 or 1) over $GF(2)$ the same as for a regular matrix, but with all arithmetic that determines linear independence performed over $GF(2)$. We define the **range** of an $n \times n$ 0-1 matrix A by

$$R(A) = \{y : y = Ax \text{ for some } x \in S_n\},$$

so that $R(A)$ is the set of numbers in S_n that are produced by multiplying each value $x \in S_n$ by A .

a. If r is the rank of matrix A , prove that $|R(A)| = 2^r$. Conclude that A defines a permutation on S_n only if A has full rank.

For a given $n \times n$ matrix A and a given value $y \in R(A)$, we define the **preimage** of y by

$$P(A, y) = \{x : Ax = y\},$$

so that $P(A, y)$ is the set of values in S_n that map to y when multiplied by A .

b. If r is the rank of $n \times n$ matrix A and $y \in R(A)$, prove that $|P(A, y)| = 2^{n-r}$.

Let $0 \leq m \leq n$, and suppose that we partition the set S_n into blocks of consecutive numbers, where the i th block consists of the 2^m numbers $i2^m, i2^m + 1, i2^m + 2, \dots, (i+1)2^m - 1$. For any subset $S \subseteq S_n$, define $B(S, m)$ to be the set of size- 2^m blocks of S_n containing some element of S . As an example, when $n = 3$, $m = 1$, and $S = \{1, 4, 5\}$, then $B(S, m)$ consists of blocks 0 (since 1 is in the 0th block) and 2 (since both 4 and 5 belong to block 2).

c. Let r be the rank of the lower left $(n - m) \times m$ submatrix of A , that is, the matrix formed by taking the intersection of the bottom $n - m$ rows and the leftmost m columns of A . Let S be any size- 2^m block of S_n , and let $S' = \{y : y = Ax \text{ for some } x \in S\}$. Prove that $|B(S', m)| = 2^r$ and that for each block in $B(S', m)$, exactly 2^{m-r} numbers in S map to that block.

Because multiplying the zero vector by any matrix yields a zero vector, the set of permutations of S_n defined by multiplying by $n \times n$ 0-1 matrices with full rank over $GF(2)$ cannot include all permutations of S_n . Let's extend the class of permutations defined by matrix-vector multiplication to include an additive term, so that $x \in S_n$ maps to $Ax + c$, where c is an n -bit vector and addition is performed over $GF(2)$. For example, when

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

and

$$c = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

we get the following permutation $\pi_{A,c}$: $\pi_{A,c}(0) = 2$, $\pi_{A,c}(1) = 1$, $\pi_{A,c}(2) = 0$, $\pi_{A,c}(3) = 3$. We call any permutation that maps $x \in S_n$ to $Ax + c$, for some $n \times n$ 0-1 matrix A with full rank and some n -bit vector c , a **linear permutation**.

- d.** Use a counting argument to show that the number of linear permutations of S_n is much less than the number of permutations of S_n .
- e.** Give an example of a value of n and a permutation of S_n that cannot be achieved by any linear permutation. (*Hint:* For a given permutation, think about how multiplying a matrix by a unit vector relates to the columns of the matrix.)

Appendix notes

Linear-algebra textbooks provide plenty of background information on matrices. The books by Strang [422, 423] are particularly good.

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